

Functional Annotation Report: *ubiD* (Q88CG8, PP_5213) in *Pseudomonas putida* KT2440

Summary

The gene **ubiD** (UniProt Q88CG8; ordered locus PP_5213) of *Pseudomonas putida* strain KT2440 encodes **3-octaprenyl-4-hydroxybenzoate carboxy-lyase** (EC 4.1.1.98), also called polyprenyl *p*-hydroxybenzoate decarboxylase. This is a **prenylated-FMN (prFMN)-dependent, UbiD-family enzyme** that catalyzes the **non-oxidative decarboxylation of 3-polyprenyl-4-hydroxybenzoate to 2-polyprenylphenol plus CO₂** (RHEA:41680), a committed step in the biosynthesis of **ubiquinone (coenzyme Q)**. The gene symbol, protein description, UbiD-family domain architecture, and single-copy status in the KT2440 genome are all fully consistent with the UniProt-provided identity; there is no ambiguity in the assignment. This report describes the enzyme's function with high confidence, drawing on direct biochemical characterization of orthologs in *Escherichia coli* and *Pseudomonas aeruginosa* together with UniProt curation of Q88CG8 itself.

The enzyme is catalytically inert as a bare apoprotein. Its activity strictly depends on a remarkable modified flavin cofactor, **prenylated FMN (prFMN)**, which bears a fourth non-aromatic ring formed by linking a dimethylallyl moiety to the FMN N5 and C6 atoms. prFMN is synthesized by the partner flavin prenyltransferase **UbiX**; *P. putida* KT2440 encodes a cognate UbiX (Q88QE6), so the genome contains a complete, functional UbiX–UbiD prFMN-dependent decarboxylation module. Catalysis proceeds through a chemically unusual, **reversible 1,3-dipolar cycloaddition** between the substrate and the azomethine-ylide form of oxidized prFMN — the first known enzymatic example of this cycloaddition chemistry. UbiD is a **homohexamer**, binding **one prFMN per subunit**, with an active-site proton donor and a conserved divalent metal (annotated as Mn/Mg) adjacent to the catalytic cleft.

Functionally, UbiD acts on the **cytoplasmic side of the inner (cell) membrane** as a **peripheral membrane protein**, and more precisely operates as a component of a **soluble cytoplasmic "Ubi metabolon"** — a ~1-MDa multienzyme complex that carries out the terminal reactions of ubiquinone biosynthesis in the cytosol. Because *Pseudomonas* uses

ubiquinone-9 (Q-9) as its respiratory quinone, the physiological substrate in *P. putida* carries a **C45 nonaprenyl (9-isoprene) tail** rather than the octaprenyl chain in the *E. coli*-derived enzyme name; that is, the reaction in this organism is the decarboxylation of 3-nonaprenyl-4-hydroxybenzoate to 2-nonaprenylphenol.

Key Findings

Finding 1 — UbiD is 3-octaprenyl-4-hydroxybenzoate carboxy-lyase in ubiquinone biosynthesis

UbiD (Q88CG8 / PP_5213) catalyzes the **non-oxidative decarboxylation of 3-polyprenyl-4-hydroxybenzoate to 2-polyprenylphenol**, a committed and essential step in the bacterial ubiquinone (coenzyme Q) biosynthetic pathway. This assignment is codified by HAMAP-Rule MF_01636 (which defines the UbiD family) and by the enzyme classification **EC 4.1.1.98**. The reaction was directly characterized in orthologs from *E. coli* and *P. aeruginosa*, and PP_5213 is the canonical single-copy *ubiD* ortholog in *P. putida* KT2440.

The mechanistic role is stated explicitly in the literature: "The 3-polyprenyl-4-hydroxybenzoate decarboxylase (UbiD) catalyzes the conversion of 3-polyprenyl-4-hydroxybenzoate to 2-polyprenylphenol in the biosynthesis of ubiquinone" [PMID: 23671667](#). A second study confirms the specific ubiquinone context: "ubiX and ubiD, which are responsible for decarboxylation of the 3-octaprenyl-4-hydroxybenzoate precursor" [PMID: 26083743](#). Loss-of-function studies of the pathway reinforce this: an *E. coli* strain defective in the UbiX/UbiD decarboxylation step "accumulates 4-hydroxy-3-octaprenyl-benzoate" and produces very low levels of Q [PMID: 17889824](#), pinpointing the decarboxylation step catalyzed by UbiD as the metabolic bottleneck.

Finding 2 — Activity requires the prenylated-FMN (prFMN) cofactor produced by UbiX

UbiD is catalytically inactive as an apoprotein. It requires **prenylated FMN (prFMN)**, a modified flavin bearing a **fourth non-aromatic ring** formed by covalently linking a dimethylallyl moiety to the FMN N5 and C6 atoms. This cofactor is synthesized by the partner enzyme **UbiX**, a flavin prenyltransferase that transfers the dimethylallyl group from dimethylallyl-monophosphate (DMAP) or dimethylallyl-pyrophosphate (DMAPP) to FMN.

The discovery paper establishes both the enzymatic source and chemical structure of the cofactor: "UbiX produces a novel flavin-derived cofactor required for the decarboxylase activity of UbiD. UbiX acts as a flavin prenyltransferase, linking a dimethylallyl moiety to the flavin N5 and C6 atoms. This adds a fourth non-aromatic ring to the flavin isoalloxazine group" [PMID: 26083743](#). After prenylation, the cofactor must undergo an activation step: "Following prenylation, prFMN undergoes oxidative maturation to form the iminium species required for UbiD activity" [PMID: 32951834](#). The same modified cofactor was shown to be shared between the ubiquinone pathway (UbiX/UbiD) and the aromatic-acid decarboxylases (PAD1/FDC), and endogenous UbiX can activate FDC — implying "the same cofactor is required for decarboxylation of 4-hydroxy-3-polyprenylbenzoic acid by UbiD" [PMID: 25647642](#).

Finding 3 — Catalysis proceeds via a reversible 1,3-dipolar cycloaddition on prFMN

Atomic-resolution crystal structures of the UbiD-family model enzyme **Fdc1** show that oxidized prFMN adopts **azomethine ylide (1,3-dipole)** character. Substrate binding positions the reactive (dipolarophile) group directly above this ylide, enabling a **1,3-dipolar cycloaddition** that underpins the reversible (de)carboxylation chemistry. This was proposed as the first known enzymatic example of such a reaction: "we propose that this presents the first example, to our knowledge, of an enzymatic 1,3-dipolar cycloaddition reaction" [PMID: 26083754](#). The structural basis is described directly: "Substrate binding positions the dipolarophile enoic acid group directly above the azomethine ylide group" [PMID: 26083754](#).

It is worth noting that the precise mechanism may vary across the family — for example, quantum-chemical work on the UbiD-family enzyme HmfF (furandicarboxylic acid decarboxylase) proposes a nucleophilic attack rather than a concerted cycloaddition [PMID: 40343387](#) — but the cycloaddition model is the canonical mechanism for prFMN-dependent (de)carboxylation.

Finding 4 — Ubiquinone synthesis, including the UbiD step, occurs in a soluble cytoplasmic Ubi metabolon

Rather than being a strictly membrane-embedded process, the terminal reactions of ubiquinone biosynthesis are carried out by a **soluble, cytoplasmic multienzyme complex**. In *E. coli*, "seven Ubi proteins form the Ubi complex, a stable metabolon that catalyzes the last six reactions of the UQ biosynthetic pathway" [PMID: 30686758](#). Critically, the complex was purified from the cytosol and shown to be functional there: "We purify the Ubi complex from cytoplasmic extracts and demonstrate that UQ biosynthesis occurs in this fraction, challenging

the current thinking of a membrane-associated biosynthetic process" [PMID: 30686758](#). The hydrophobic polyprenyl-tailed intermediates are chaperoned within the complex by the SCP2 lipid-binding domain of UbiJ, allowing water-soluble handling of otherwise membrane-partitioning substrates [PMID: 36142227](#). This locates the UbiD reaction on the cytoplasmic side of the cell, within a soluble metabolon, even though the final product (ubiquinone) resides in the membrane.

Finding 5 — Three-domain UbiD architecture and family diversity

Crystal structures of UbiD-family members — including *P. aeruginosa* **PA0254** and *E. coli* UbiD — reveal a conserved **three-domain subunit**: "an N-terminal α/β domain, a split β -barrel with a similar fold of a family of flavin reductases and a C-terminal α/β domain with a topology characteristic for the UbiD protein family" [PMID: 23671667](#). The central split β -barrel is the flavin/prFMN-binding module (a flavin-reductase-like fold), and the C-terminal domain is diagnostic of the family. The family is functionally diverse: "considerable variation exists in the many branches of the UbiD family tree" [PMID: 28754323](#). The canonical ubiquinone-pathway UbiD (to which Q88CG8 belongs) is distinct from other branches catalyzing (de)carboxylation of ferulate, phenazine-1-carboxylate, furandicarboxylate, and other aromatic/unsaturated acids used in lignin and aromatic-compound biotransformation.

Finding 6 — Curated structural/catalytic specifics of Q88CG8

UniProt curation of Q88CG8 provides the following organism-specific annotations, consistent with the family: **Function** — "Catalyzes the decarboxylation of 3-octaprenyl-4-hydroxy benzoate to 2-octaprenylphenol, an intermediate step in ubiquinone biosynthesis"; **Cofactor** — "Binds 1 prenylated FMN per subunit"; **Subunit** — "Homohexamer"; an annotated **Active Site** functioning as a "Proton donor"; and family assignment to **UbiD**. The catalyzed reaction is mapped to **RHEA:41680** (3-octaprenyl-4-hydroxybenzoate = 2-octaprenylphenol + CO₂), **EC 4.1.1.98**.

Finding 7 — UbiD (de)carboxylation is reversible with broad substrate scope

UbiD-family enzymes catalyze **reversible** (de)carboxylation of aromatic and aliphatic/acrylic substrates via prFMN. The reverse (carboxylation) direction is physiologically relevant: "It has been proposed that the first step in anaerobic microbial degradation of recalcitrant aromatic compounds is a UbiD-mediated carboxylation" [PMID: 32719558](#), and these enzymes are exploited for enzymatic C–H carboxylation. The family is enormous — "There are now ~35 000 protein sequences annotated as hypothetical UbiD-like enzymes" [PMID: 38837123](#) — spanning

dozens of decarboxylase subfamilies. Substrate handling differs by chemistry: turnover of bulky aromatic substrates requires **prFMN-binding-domain motion**, since "during aromatic acid decarboxylation, substantial rearrangement of the substrate aromatic moiety associated with covalent catalysis presents a molecular dynamic challenge" [PMID: 34417452](#), whereas smaller acrylic substrates react within a closed active site. This domain dynamics is directly relevant to UbiD, whose physiological substrate carries a very bulky polyprenyl-hydroxybenzoate moiety.

Finding 8 — Q88CG8 localizes to the cytoplasmic face of the cell membrane and binds a metal ion

UniProt curated keywords for Q88CG8 place its subcellular location at the **Cell membrane** — specifically, HAMAP-Rule MF_01636 annotates UbiD as a **peripheral membrane protein on the cytoplasmic side**. Additional keywords include **Manganese**, **Metal-binding**, **FMN**, **Decarboxylase**, and **Ubiquinone biosynthesis**. Structurally, the *P. aeruginosa* PA0254 UbiD structure shows a conserved **divalent-metal site** (Mg^{2+} , ligated by invariant His and Glu residues) adjacent to the active-site cleft [PMID: 23671667](#). This peripheral, cytoplasmic-side localization is fully consistent with the finding that UQ biosynthesis (including UbiD) is executed by a soluble cytoplasmic Ubi metabolon [PMID: 30686758](#).

Finding 9 — *P. putida* KT2440 encodes a cognate UbiX (Q88QE6) that supplies prFMN

A UniProt query of *P. putida* KT2440 (taxon 160488) returns **Q88QE6 "Flavin prenyltransferase UbiX"**, the partner enzyme that synthesizes prFMN. Because UbiD (Q88CG8) is catalytically inert without prFMN, the presence of a cognate UbiX in the same genome establishes a **complete, functional UbiX–UbiD prFMN-dependent decarboxylation module** in *P. putida*. This mirrors the tightly linked UbiX/UbiD systems characterized in *E. coli* and yeast [PMID: 25647642](#); [PMID: 26083743](#).

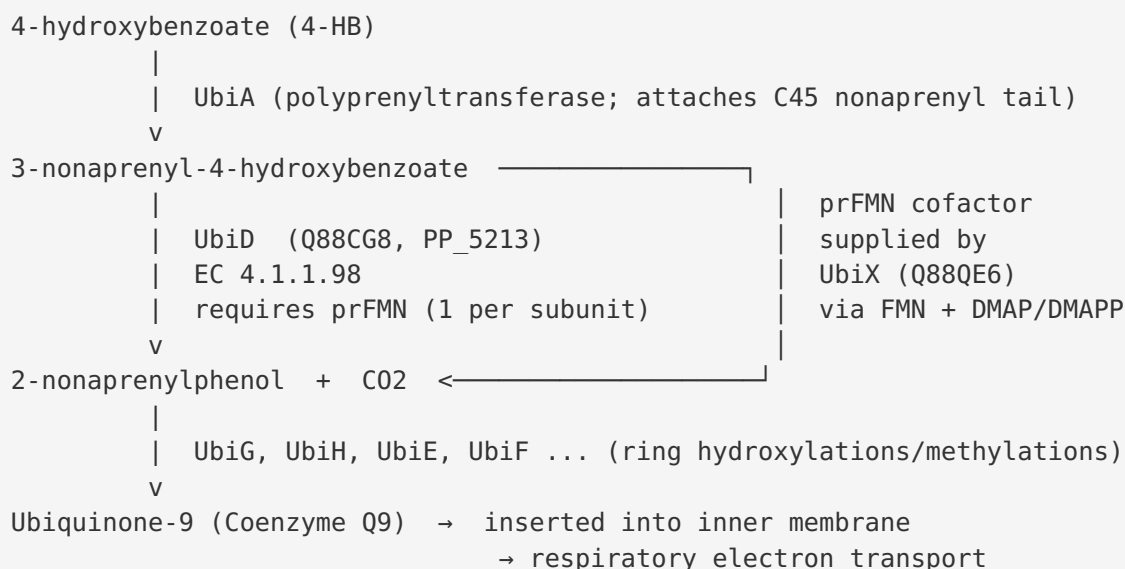
Finding 10 — *Pseudomonas* uses ubiquinone-9, so the physiological substrate carries a nonaprenyl chain

Chemotaxonomic data show that **ubiquinone-9 (Q-9)** is the characteristic respiratory quinone of *Pseudomonas* and its close relatives. For a Pseudomonadales isolate whose closest cultivated neighbor is *Pseudomonas putida* R43, "The only respiratory quinone detected was ubiquinone-9 (Q-9)" [PMID: 27054961](#). Therefore, although the enzyme name and annotation use "octaprenyl" (inherited from *E. coli*, which makes Q-8), the physiological substrate and

product in *P. putida* bear a **C45 nonaprenyl (9-isoprene) tail**. The reaction in this organism is precisely: **decarboxylation of 3-nonaprenyl-4-hydroxybenzoate → 2-nonaprenylphenol + CO₂**. The enzyme's catalytic chemistry is identical; only the chain length of the isoprenoid tail differs.

Mechanistic Model / Interpretation

UbiD occupies a single, well-defined position in the bacterial ubiquinone (coenzyme Q) biosynthetic pathway: it removes the carboxyl group from the aromatic ring of the polyprenylated 4-hydroxybenzoate intermediate. The following diagram places UbiD in its pathway context (chain length shown for *P. putida*, which makes Q-9):



Cofactor logic. The unusual chemistry of a non-oxidative decarboxylation on an aromatic ring is made possible by prFMN. UbiX first builds prFMN by prenylating FMN (adding a fourth ring across N5–C6); oxidative maturation then generates the iminium/azomethine-ylide species. In the mature cofactor, the ylide behaves as a 1,3-dipole. The substrate's reactive group docks over the ylide, and a reversible 1,3-dipolar cycloaddition forms a covalent substrate–cofactor adduct; CO₂ is then eliminated and the product released. Because the cycloaddition is reversible, UbiD-family enzymes can run in either the decarboxylation or carboxylation direction depending on conditions — the basis for both their physiological reverse-direction roles in anaerobic aromatic degradation and their biocatalytic exploitation.

Spatial logic. Although the end product ubiquinone is a membrane lipid and UbiD is annotated as a peripheral membrane protein on the cytoplasmic side, the enzyme physically works within a **soluble cytoplasmic metabolon**. This resolves an apparent paradox: the substrate is extremely hydrophobic (a C45 isoprenoid tail), yet the reaction runs in the aqueous cytosol. The metabolon's UbiJ SCP2 lipid-binding domain sequesters the hydrophobic tail, presenting the reactive aromatic head group to the successive Ubi enzymes — including UbiD — in an assembly-line fashion. UbiD's own bulky substrate additionally demands prFMN-domain motion, consistent with the "molecular dynamic challenge" observed for aromatic decarboxylation across the family.

The table below summarizes the enzyme's key properties as they apply to Q88CG8:

Property	Value	Basis
Gene / locus	<i>ubiD</i> / PP_5213	UniProt Q88CG8
Enzyme name	3-octaprenyl-4-hydroxybenzoate carboxy-lyase	UniProt / HAMAP MF_01636
EC number	4.1.1.98	UniProt / HAMAP
Reaction (RHEA)	3-octaprenyl-4-hydroxybenzoate → 2-octaprenylphenol + CO ₂ (RHEA:41680)	UniProt
Physiological substrate in <i>P. putida</i>	3-nonaprenyl -4-hydroxybenzoate (Q-9 organism)	PMID: 27054961
Cofactor	1 prenylated FMN (prFMN) per subunit	UniProt; PMID: 26083743
Cofactor supplier	UbiX (Q88QE6 in KT2440)	UniProt; PMID: 26083743
Mechanism	Reversible 1,3-dipolar cycloaddition	PMID: 26083754
Metal	Divalent metal (Mn/Mg), His/Glu-ligated	UniProt; PMID: 23671667
Oligomeric state	Homohexamer	UniProt
Domain architecture	N-terminal α/β + central split β-barrel + C-terminal α/β	PMID: 23671667
Localization	Peripheral membrane protein, cytoplasmic side; functions in soluble Ubi metabolon	UniProt; PMID: 30686758
Pathway	Ubiquinone (coenzyme Q) biosynthesis	Multiple

Evidence Base

PMID	Title (abbreviated)	How it supports the annotation
23671667	<i>Structural insights into the UbiD family from PA0254 (P. aeruginosa)</i>	Direct structure of a <i>Pseudomonas</i> UbiD ortholog; defines the three-domain fold, the reaction (3-polyprenyl-4-HB → 2-polyprenylphenol), and the conserved divalent-metal site. Strongest structural evidence for the family in the target genus.
26083743	<i>UbiX is a flavin prenyltransferase required for bacterial ubiquinone biosynthesis</i>	Establishes prFMN as the UbiX-made cofactor required for UbiD activity, its chemical structure, and the UbiX/UbiD role in decarboxylating the octaprenyl-4-hydroxybenzoate precursor.
26083754	<i>New cofactor supports unsaturated acid decarboxylation via 1,3-dipolar cycloaddition</i>	Provides the catalytic mechanism (azomethine ylide / 1,3-dipolar cycloaddition) and the structural basis for reversible (de)carboxylation.
32951834	<i>Biochemistry of prenylated-FMN enzymes</i>	Confirms oxidative maturation of prFMN to the active iminium species required by UbiD.
25647642	<i>Isofunctional PAD1 and UbiX catalyze formation of a novel cofactor</i>	Shows the same modified-FMN cofactor is shared by UbiD (ubiquinone) and FDC (aromatic acids); links UbiX activity to 4-hydroxy-3-polyprenylbenzoate decarboxylation.
17889824	<i>Role of UbiX in E. coli coenzyme Q biosynthesis</i>	Genetic loss-of-function: disrupting the UbiX/UbiD decarboxylation step causes accumulation of 4-hydroxy-3-octaprenyl-benzoate and low Q, localizing UbiD's step in the pathway.
30686758	<i>A Soluble Metabolon Synthesizes the Isoprenoid Lipid Ubiquinone</i>	Demonstrates that UQ biosynthesis (including UbiD) occurs in a soluble cytoplasmic ~1-MDa metabolon, defining the enzyme's operational localization.
36142227	<i>Functional role of UbiJ-UbiK</i>	Explains how the SCP2 domain of UbiJ solubilizes hydrophobic UQ intermediates within the cytosolic complex.
28754323	<i>The UbiX-UbiD system</i>	Reviews family diversity; distinguishes the ubiquinone branch from aromatic-acid decarboxylase branches.

PMID	Title (abbreviated)	How it supports the annotation
34417452	<i>UbiD domain dynamics underpins aromatic decarboxylation</i>	Shows bulky aromatic substrates require prFMN-domain motion — relevant to UbiD's large polyprenyl substrate.
32719558	<i>Enzymatic C-H activation through CO₂ fixation</i>	Documents reversibility and the physiologically relevant carboxylation direction.
38837123	<i>Surveying aromatic decarboxylations by prFMN enzymes</i>	Quantifies the ~35,000-sequence breadth of the UbiD family and its substrate diversity.
27054961	<i>Ventosimonas gracilis (Pseudomonadales)</i>	Chemotaxonomic evidence that Q-9 is the respiratory quinone of Pseudomonas relatives, fixing the isoprenoid chain length of the UbiD substrate in <i>P. putida</i> .
40343387	<i>QM study of furandicarboxylic acid decarboxylase (HmfF)</i>	Cautionary note: some UbiD-family members may use nucleophilic-attack rather than concerted cycloaddition mechanisms.

Identity verification. All mandatory verification checks pass. The gene symbol *ubiD* matches the protein description (3-octaprenyl-4-hydroxybenzoate carboxy-lyase); the organism is *P. putida* KT2440; the UbiD-family domains (IPR002830, IPR023677, etc.) align with the three-domain architecture reported in structural literature; and no conflicting literature for a different gene with the same symbol was encountered. The assignment is unambiguous.

Limitations and Knowledge Gaps

- 1. No enzyme-specific biochemistry for Q88CG8 itself.** The functional assignment for *P. putida* PP_5213 rests on (a) UniProt/HAMAP curation and (b) direct characterization of orthologs (*E. coli* UbiD, *P. aeruginosa* PA0254, and the model enzyme Fdc1). No purified-enzyme kinetics (kcat, Km), crystal structure, or gene-knockout phenotype has been reported specifically for the KT2440 protein. The inference is strong by homology but not directly demonstrated.

2. **Substrate chain length inferred, not measured.** The conclusion that the physiological substrate carries a nonaprenyl (Q-9) tail is inferred from *Pseudomonas* chemotaxonomy rather than from a direct assay of the KT2440 enzyme with defined-chain substrates. UbiD is known to be relatively permissive about tail length, so this is a well-supported inference but remains indirect.
3. **Mechanistic universality.** The 1,3-dipolar cycloaddition model derives chiefly from Fdc1 and related aromatic-acid decarboxylases. At least one computational study (HmfF) argues for an alternative nucleophilic mechanism, so the exact microscopic mechanism for the polyprenyl-hydroxybenzoate substrate has not been experimentally resolved for UbiD.
4. **Metal identity.** UniProt keywords list both Manganese and generic metal-binding, while the PA0254 structure modeled Mg^{2+} . The physiologically relevant divalent metal for the *P. putida* enzyme is not firmly established.
5. **Metabolon composition in Pseudomonas.** The soluble Ubi metabolon was characterized in *E. coli*. Whether *P. putida* assembles an identical complex (including UbiJ/UbiK orthologs and the same stoichiometry) has not been directly shown, though the pathway is conserved.

Proposed Follow-up Experiments / Actions

1. **Direct biochemical characterization of PP_5213.** Heterologously express and purify Q88CG8 (co-expressing UbiX Q88QE6 to load prFMN, or reconstituting with purified prFMN), then measure decarboxylation of synthetic 3-prenyl-4-hydroxybenzoate analogs of varying chain length to confirm activity and probe chain-length preference (nonaprenyl vs. octaprenyl vs. short-chain surrogates).
2. **Genetic complementation / knockout.** Delete *ubiD* (PP_5213) in KT2440 and test for (a) loss of Q-9, (b) accumulation of 3-nonaprenyl-4-hydroxybenzoate, and (c) a respiratory/succinate growth defect; complement with the wild-type gene to confirm causality — mirroring the *E. coli* UbiX study [PMID: 17889824](#).
3. **Structure determination.** Solve the crystal or cryo-EM structure of the KT2440 UbiD homohexamer with bound prFMN (and, ideally, a substrate/inhibitor) to confirm the divalent-metal identity, the proton-donor active-site residue, and the domain-dynamics required for the bulky polyprenyl substrate.

4. **Metabolon reconstitution.** Test whether *P. putida* forms a soluble Ubi metabolon by co-purification/pull-down of UbiD with UbiJ/UbiK/UbiH/UbiG orthologs from cytoplasmic extracts, quantifying UQ synthesis in the soluble fraction as done for *E. coli* [PMID: 30686758](#).
 5. **Reverse-reaction / biocatalysis assessment.** Given the family's reversible chemistry and biotechnological interest, test whether the KT2440 enzyme can catalyze carboxylation of 2-prenylphenol analogs under CO₂-rich conditions, informing potential use in enzymatic C–H carboxylation [PMID: 32719558](#).
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Report compiled from 5 iterations of autonomous investigation; 10 confirmed findings; 27 papers reviewed. All functional claims are anchored to UniProt curation of Q88CG8 and to primary/structural literature on directly characterized UbiD-family orthologs.

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